Week 5 – False Positive Analysis

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Target Alert Metadata:

Alert Name:	New Account Creation – Linux useradd event	**Observed Account:**	`splunk` (UID: 1001, GID: 1001)
Target Host:	`ubuntu-soc`	**Timestamp:**	2026-11-05 12:40:11.545 IST
Observed Details:	Home Directory: `/opt/splunk` | Shell: `/bin/bash` | Session: `/dev/pts/1`		

1. ALERT SUMMARY

This security alert triggered upon detecting the execution of a Linux account creation command (`useradd`) on host `ubuntu-soc` . In security operations monitoring, new user account creation represents a critical system state change that maps to **MITRE ATT&CK T1136 (Create Account)** as a primary persistence mechanism. The captured alert logged the precise timestamp (`2026-11-05 12:40:11.545 IST`), created account name (`splunk`), assigned UID 1001, GID 1001, home path (`/opt/splunk`), interactive shell (`/bin/bash`), and terminal session origin (`/dev/pts/1`).

2. WHY IT WAS CONSIDERED A FALSE POSITIVE

The account creation event is highly consistent with standard administrative procedures involved in installing and deploying Splunk Enterprise within the local lab environment. Specifically, the username matches the software application (`splunk`), the designated home directory (`/opt/splunk`) aligns with default Splunk installation paths, and the terminal origin (`/dev/pts/1`) reflects active administrative installation commands.

Crucial Distinction: This evaluation is based on contextual analysis of log attributes matching expected deployment steps. It must be clearly distinguished from independently verified cryptographic proof or signed change ticket validation, as adversaries could potentially mask backdoor accounts using naming conventions of known applications.

3. INVESTIGATION PERFORMED

The analyst reviewed available log fields within the `linux_secure` index, inspecting the raw event payload (`_raw`), timestamp (`_time`), target host (`ubuntu-soc`), and process ID (`useradd[6140]`). The evidence confirms that parameters match standard application setup rather than anomalous privilege escalation.

To achieve full confirmation and rule out residual risk, the following additional verification steps are required:

- Cross-referencing terminal history and preceding/succeeding process logs (such as `sudo` commands) to verify the initiating user.
- Checking lab deployment documentation or administrative change records matching the exact execution timestamp.
- Verifying parent process execution trees to ensure `useradd` was directly spawned by an authorized administrator terminal.

4. POSSIBLE DETECTION TUNING

To prevent recurring false positives without disabling essential account creation visibility, detection rules must be tuned thoughtfully. SOC analysts must **never** implement blanket suppression of all `useradd` events or permanently whitelist the `splunk` account name. Instead, rules should incorporate contextual filters, such as matching account creations against approved software baseline lookups, inspecting parent process paths, and differentiating interactive terminal executions from unauthenticated script triggers.

5. DETECTION ENGINEERING RECOMMENDATIONS

- **Context-Aware Rules:** Develop SIEM alert logic correlating `useradd` execution with active administrative session IDs and approved deployment windows.
- **Approved Baseline Lookups:** Maintain an authorized service account lookup table (e.g., `splunk`, `wazuh`, `nginx`) to dynamically flag non-standard account creation.
- **Parent Process & User Tracking:** Enrich alerts by capturing parent process IDs (PPID), command lines, and initiating user contexts to trace command origin.
- **Risk-Based Alert Severity:** Assign low/medium severity to matched service account creations while reserving high severity for unauthorized root additions or non-standard execution hours.

6. CONCLUSION

This analysis evaluated a Linux `useradd` alert on host `ubuntu-soc` for user `splunk`. While contextual evidence strongly indicates legitimate software installation activity, the investigation underscores the necessity of validating all alert parameters prior to closure. For entry-level SOC analysts, thorough verification and balanced detection tuning remain fundamental to maintaining strong security posture and avoiding premature alert dismissal.